Weak Acids and Ka

Answer key with worked solutions

A. Understanding

1. Ka expression.

> $K_a = \frac{[H_3O^+][A^-]}{[HA]}$. A large K_a means the acid ionizes to a greater extent: it is a stronger acid.

2. Ranking and pKa of HF.

> Strongest to weakest: **HF > acetic acid > HCN** (largest to smallest Ka). pKa(HF) = −log(6.8 × 10^{-4}) = **3.17**

B. pH of weak acids

3. 0.20 M acetic acid.

> $[H_3O^+] \approx \sqrt{1.8 \times 10^{-5} \times 0.20} = 1.9 \times 10^{-3}$ M (about 0.9% of 0.20 M, so the shortcut is valid). pH = **2.72**; percent ionization ≈ **0.9%**

4. 0.25 M HCN.

> $[H_3O^+] \approx \sqrt{4.9 \times 10^{-10} \times 0.25} = 1.1 \times 10^{-5}$ M. pH = **4.96**

C. Ka from a measured pH

5. 0.050 M acid, pH 3.20.

> $[H_3O^+] = [A^-] = 10^{-3.20} = 6.3 \times 10^{-4}$ M; [HA] = 0.050 M − 0.00063 M = 0.0494 M. $K_a = (6.3 \times 10^{-4})^2 \div 0.0494 =$ **8.1×10^{-6}**; percent ionization = **1.3%**

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D. Weak bases

6. 0.20 M ammonia.

> $[\mathrm{OH^-}] \approx \sqrt{1.8 \times 10^{-5} \times 0.20} = 1.9 \times 10^{-3}$ M; pOH = 2.72; pH = **11.28**

7. Kb of acetate.

> $K_\mathrm{b} = K_\mathrm{w} / K_\mathrm{a} = 1.0 \times 10^{-14} \div 1.8 \times 10^{-5} = \mathbf{5.6 \times 10^{-10}}$

E. Challenge

8. Why the pH values differ.

> HCl is a **strong** acid: it ionizes completely, so [H₃O⁺] = 0.20 M and pH = 0.70. Acetic acid is **weak**: less than 1% ionizes, so [H₃O⁺] is only 1.9×10^{-3} M and pH = 2.72.

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